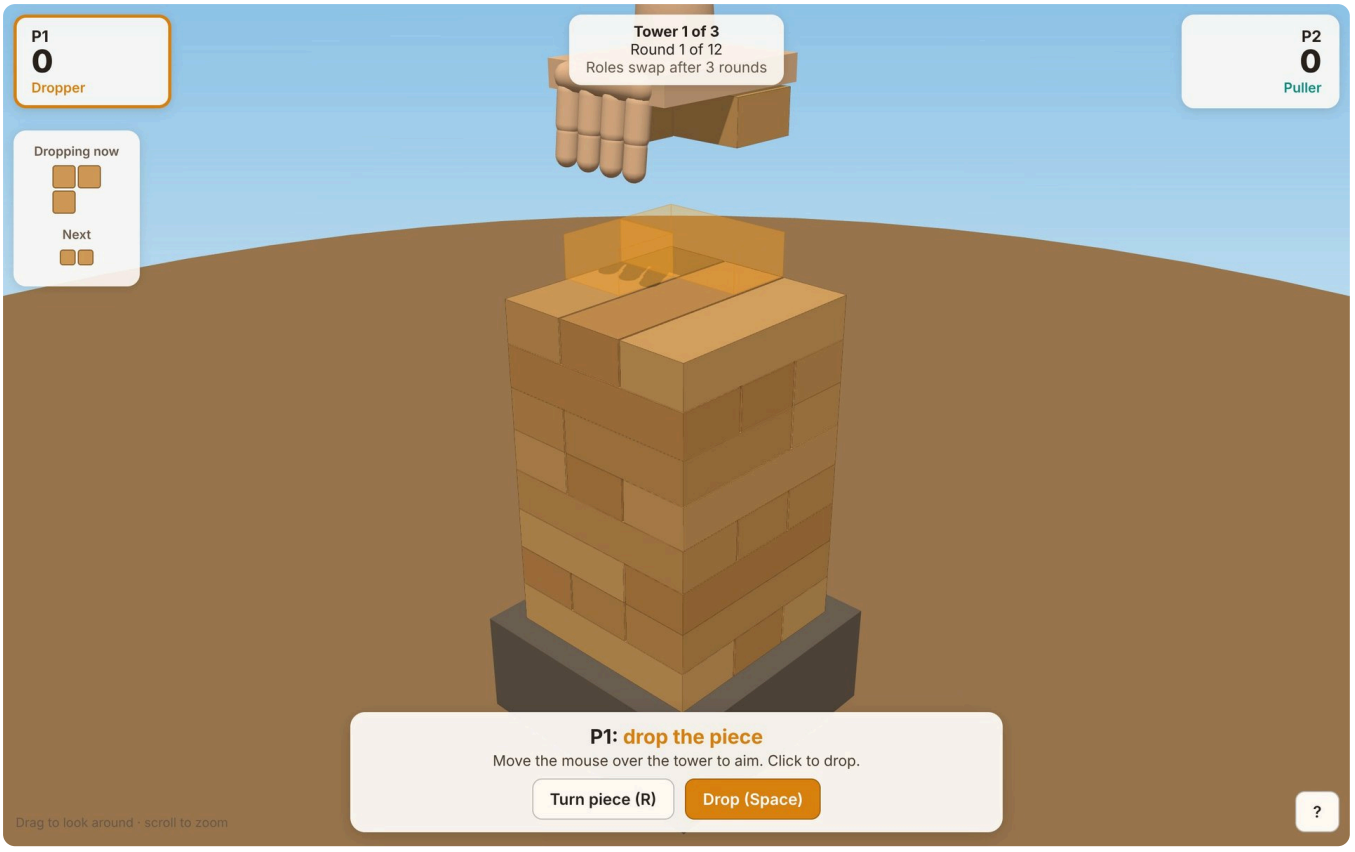
Counterweight

Player's guide · turn-based 3D prototype



Counterweight is a two-player tower game. Both players share one wooden tower and take turns. One **drops** a piece on top. The other **pulls** a block out. Every few rounds you swap jobs. Don't be the one who knocks it down.

This guide covers the turn-based 3D version. It is an early test build, so expect rough edges. There is no sound yet and no computer opponent.

1. Starting the game

You need a Windows PC or a Mac with a mouse, the game folder, and Python installed. The game runs in your web browser, loaded through a small local server.

> **Already have an older copy?** This version replaces the earlier 2D one. Download the ZIP again, unzip it, and use the new `counterweight` folder.

Windows

1. Open the game folder in File Explorer and go into the `counterweight` folder.
2. Click the address bar at the top of the window, type `cmd` and press Enter. A black window opens.
3. Type this and press Enter:

   ```
   python -m http.server 8420
   ```

4. The window shows `Serving HTTP on :: port 8420` and then sits there. That means it is running. Leave it open.
5. Open Chrome, Edge or Firefox and go to `http://localhost:8420/`

Mac

1. Open the Terminal app. Type `cd` and a space, drag the `counterweight` folder onto the window, and press Enter.
2. Type `python3 -m http.server 8420` and press Enter. Leave the window open.
3. Open your browser and go to `http://localhost:8420/`

When the menu appears, click **Start game**. To stop the server when you are done, close the black window.

2. The game in one minute

- Two players, P1 and P2, share one tower and one mouse. You take turns; there is no clock.
- A **round** has two moves. First the **Dropper** places a piece on top. Then the **Puller** slides one block out.
- After every **3 rounds** you swap roles.
- A tower lasts **12 rounds**, or until it falls.
- If the tower falls on your move, you lose 10 points.
- A game is **3 towers**. The player with the most points at the end wins.

The top of the screen shows each player's points and current role. The box around a player lights up when it is their move, and the panel at the bottom always says who should do what.

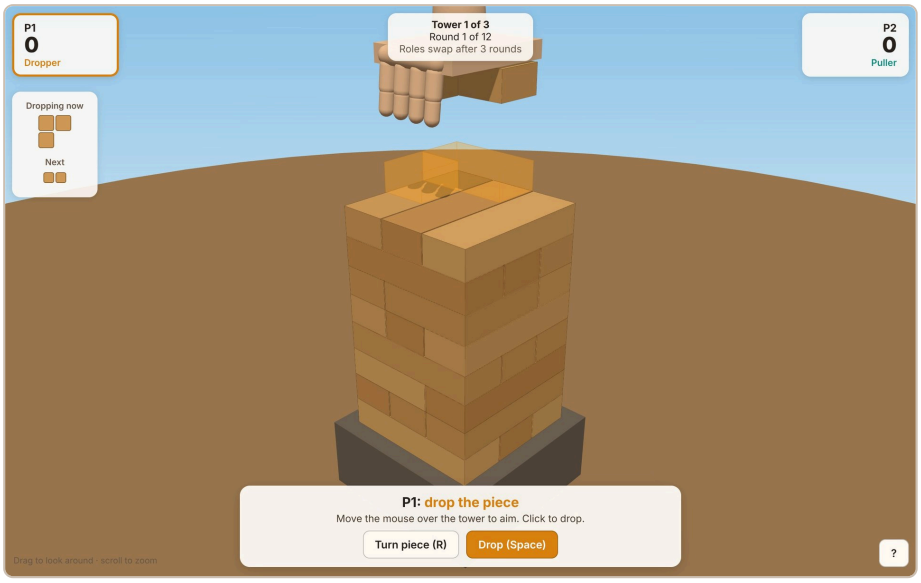
3. Looking around

The tower is fully 3D. You can look at it from any side at any time, including during the other player's move.

Do this	To
Hold the mouse button and drag	Turn the view around the tower, or tilt it up and down
Scroll the mouse wheel	Zoom in and out
Arrow keys	Turn and tilt the view in steps
The **?** button, bottom right	Show the rules

A drag never drops or pulls anything. Only a click without moving the mouse does.

4. The Dropper's move



The hand holds the piece above the tower. The orange shadow shows where it will land.

A hand comes down from the sky holding your piece. The panel on the left shows the piece you are dropping now and the one that comes next.

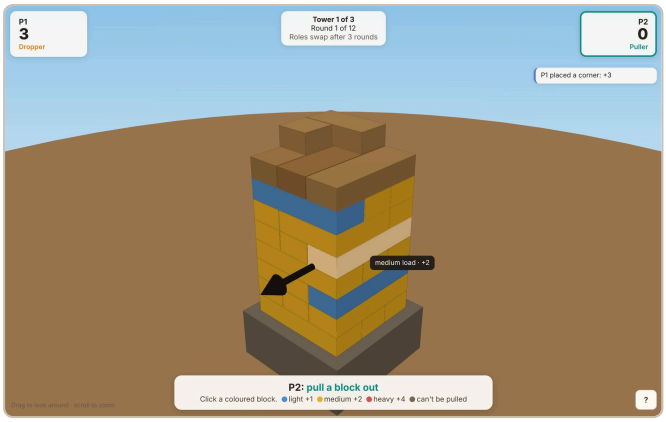
1. **Aim.** Move the mouse over the top of the tower. The hand follows, and an orange shadow shows where the piece will land.
2. **Turn the piece** if you want: press `R`, right-click, or use the **Turn piece** button.
3. **Drop.** Click, press `Space`, or use the **Drop** button. The hand lowers the piece, lets go and leaves.

The piece always stays inside the tower's 3 × 3 outline. It lands on the highest thing underneath it, so a piece that rests half on a block and half over a gap will sit tilted, or slide off.

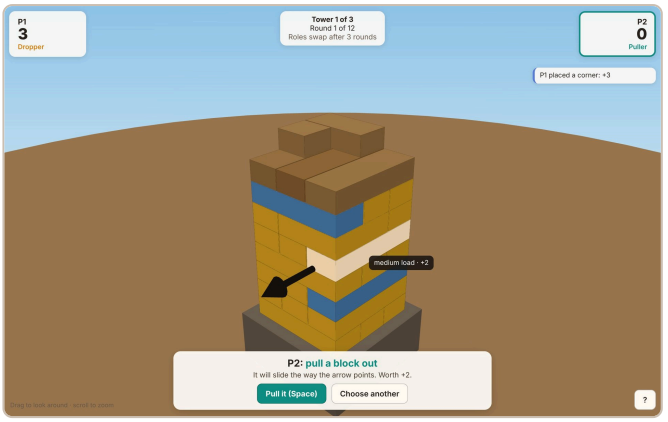
The pieces are all flat and one layer tall: the plank (3 squares in a row), the half (2 in a row), the corner (an L of 3), the slab (a 2 × 2 square) and the cube (1 square).

5. The Puller's move

When it is your turn to pull, the tower changes colour to show what you can take and what each block is worth.



Point at a block to see its load and value.



Click once to choose it. The arrow shows which way it will slide.

1. **Choose.** Click a coloured block. It lights up and an arrow shows which way it will slide out.
2. **Pull.** Click the same block again, press `Space`, or use the **Pull it** button. To change your mind first, click a different block or **Choose another**.

The block slides straight out, towards the side you are looking from. If you want it to come out the other way, turn the view first.

What the colours mean

Colour	Meaning	Points
Blue	Light load. Carrying less than its share. Safest to take.	+1
Yellow	Medium load. Carrying about its share.	+2
Red	Heavy load. Holding up more than its share. Risky.	+4
Dark brown	Can't be pulled.	

Blocks you can't pull

- Anything in the **top two layers**.
- A block that is **boxed in**: it needs a clear straight path to the outside of the tower.

Point at a dark block and the label tells you which of the two reasons applies. If nothing at all can be pulled, the pull is skipped.

Weight matters

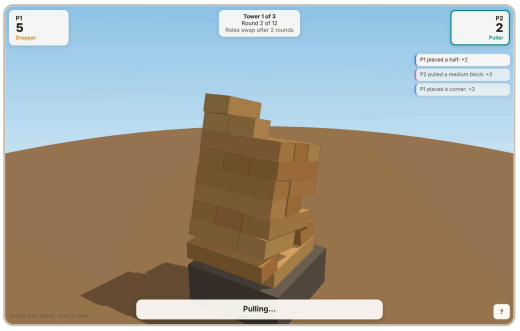
A block that is carrying a lot drags on whatever is sitting on it as it slides out. Light blocks come out cleanly. At the start nearly everything is yellow; as gaps appear, the weight shifts and red and blue blocks show up.

6. Points

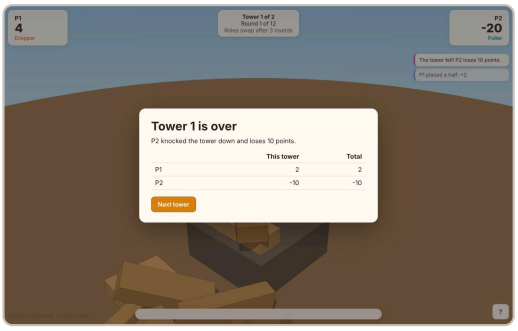
What happened	Who	Points
Placed a piece that stayed on the tower	Dropper	+1 per square
That piece completed a full 3 × 3 layer	Dropper	+3
Pulled a blue / yellow / red block	Puller	+1 / +2 / +4
Knocked blocks off the tower (one or two)	Whoever moved	−2 each
Knocked the tower down (three or more blocks fall)	Whoever moved	−10

The number under your name is your total for the whole game so far. Recent scoring is listed on the right of the screen.

7. When the tower falls



Three or more blocks hitting the table counts as a collapse.



The end-of-tower screen.

After every move the game waits for the tower to stop moving, then checks what fell. If three or more blocks ended up on the table, the tower is down: the player who made that move loses 10 points and scores nothing else for it, and the tower ends.

A new tower then starts. The players take turns to be the first Dropper: P1 on the first tower, P2 on the second, P1 on the third. After the third tower, the higher total wins.

8. Tips

As Dropper

- Look at the next piece before you place this one, and leave a gap it will fit.
- Finishing a layer is worth more than most pieces.
- You swap roles soon. A top-heavy tower is a problem for whoever pulls next.

As Puller

- Walk the view right round the tower before you choose.
- Red blocks pay four times as much as blue ones, and are four times as likely to cost you ten.
- Taking two blocks from the same side of one layer is how towers tip over.

9. Troubleshooting

Problem	What to do
The black window seems stuck on `Serving HTTP on…`	That is normal. The server is running. Leave the window open and go to the browser.
`'python' is not recognized`	Python is not installed, or was installed without the **Add python.exe to PATH** box ticked. Run the installer from python.org again and tick that box.
The browser says the site can't be reached	Check the black window is still open and the address is `http://localhost:8420/`
You see a list of files, or the Route Lens app	The server was started one folder too high. Add `counterweight/` to the end of the address.
You still see the old flat 2D game	You are running an old copy. Download the ZIP again and start the server from the new `counterweight` folder. If the address is right and it still shows, press `Ctrl`+`F5` to reload.
The page stays on "Loading..."	The page was probably opened by double-clicking the file. Open it through the `http://localhost` address instead.
Clicking drops the piece when you only wanted to look	Hold the button and move the mouse to look around. A click without movement is a drop.
The block slides out the wrong way	It comes towards the side you are viewing from. Turn the view, check the arrow, then pull.

Settings for test games

Add these to the end of the address, for example `http://localhost:8420/?rounds=6&towers=1`

Setting	Meaning	Normal value
`rounds`	Rounds per tower	12
`towers`	Towers per game	3
`seed`	Any number. The same number gives the same order of pieces each time.	random

10. Not in this version yet

Sound, a computer opponent (one person can still play both sides to try it out), online play, and a hand for the Puller. The older real-time 2D version is still linked from the start menu. Point values and round counts are starting points and may change.